

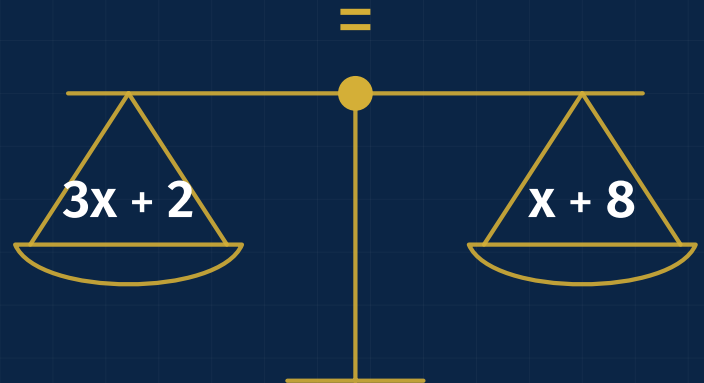


DR ESLAM AHMED
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CHAPTER 2 • ALGEBRA • TOPIC 9

Linear Equations

One method for every linear equation:
clean it, then **isolate** the letter. Eight
question types, all from real exams.



8

VARIANTS

12

PAST-PAPER QUESTIONS

37

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Letters on both sides

V2 Brackets

V3 Fraction = number

V4 Fraction = expression

V5 Letter in the bottom

V6 Two fractions

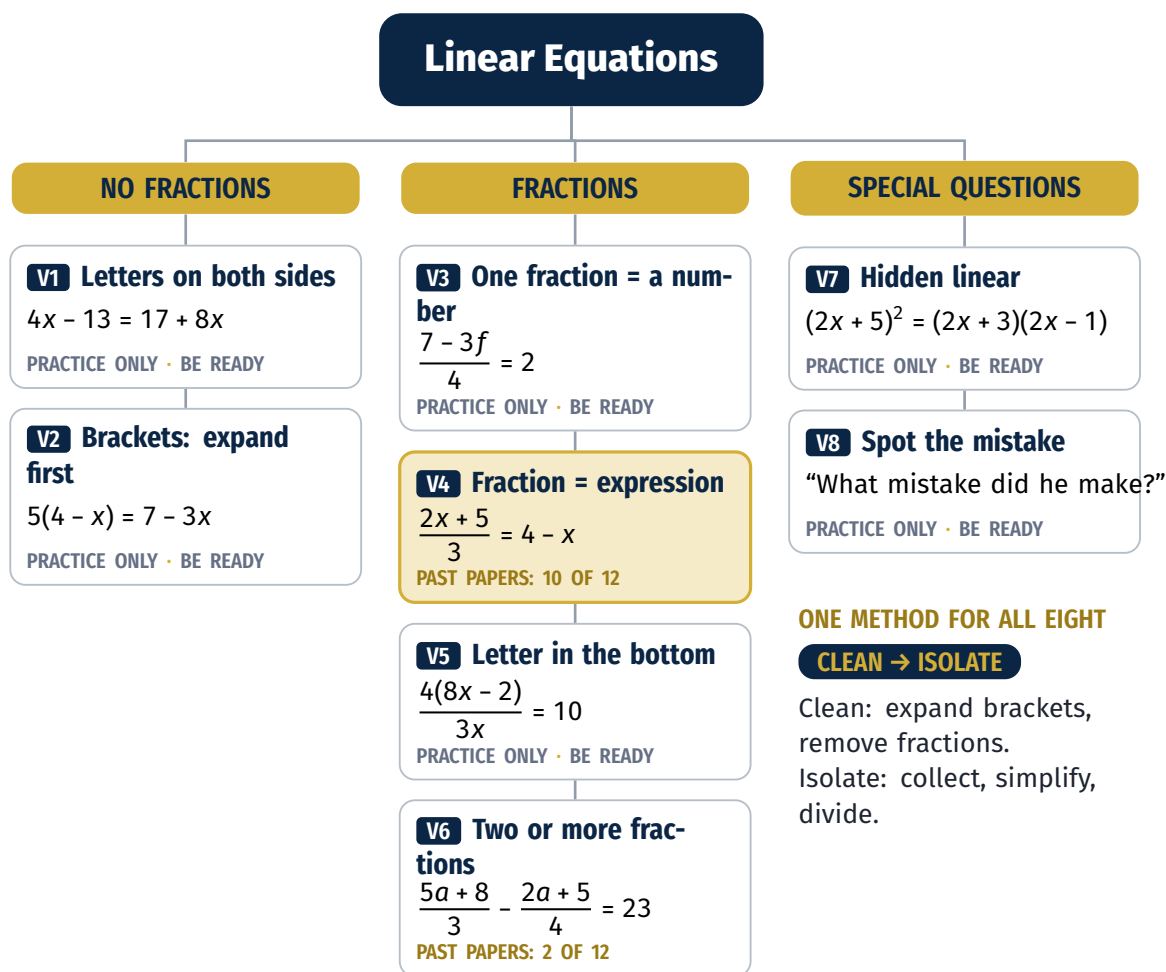
V7 Hidden linear

V8 Spot the mistake

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Linear Equations at a glance



Your notes


How often does each variant come up?

12 past-paper questions (Jan 2020 → Jun 2025) and 37 practice parts, grouped by type.

	PAST PAPERS (OUT OF 12)	PRACTICE PARTS
V1 Letters on both sides	● ● ● ● ● ● ● ● ● ● ● ● 0	■ 3
V2 Brackets: expand first	● ● ● ● ● ● ● ● ● ● ● ● 0	■ 9
V3 One fraction = a number	● ● ● ● ● ● ● ● ● ● ● ● 0	■ 2
V4 Fraction = expression	● ● ● ● ● ● ● ● ● ● ● ● 10	■ 11
V5 Letter in the bottom	● ● ● ● ● ● ● ● ● ● ● ● 0	■ 2
V6 Two or more fractions	● ● ● ● ● ● ● ● ● ● ● ● 2	■ 7
V7 Hidden linear	● ● ● ● ● ● ● ● ● ● ● ● 0	■ 2
V8 Spot the mistake	● ● ● ● ● ● ● ● ● ● ● ● 0	■ 1

WHAT THIS TELLS YOU Every past-paper linear equation had a **fraction** in it. 10 of 12 were **V4**. Master V4 and V6 first — then the rest are quick wins.

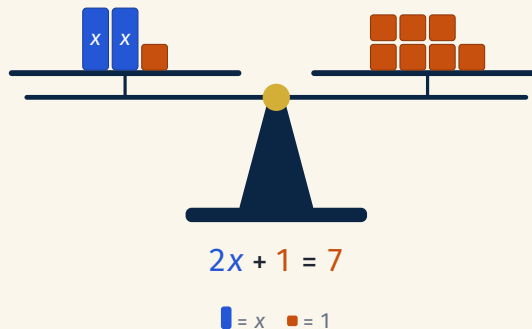
BEFORE YOU START

The big idea

**An equation is a balance.**

The two sides weigh the **same**.
Whatever you do to one side,
do it to the other side — and it stays
balanced.

In a **linear** equation the letter has **no powers**: there is x , but no x^2 . It has exactly **one answer**.



Undo with the opposite



$2x + 1 = 7$: the $+1$ is undone by -1 , the $\times 2$ is undone by $\div 2$.

Dr Eslam's method: Clean, then Isolate

**STAGE 1
CLEAN**

- Expand every bracket
 - Remove every fraction ($\times$ the bottom, or $\times$ the LCM)
- Now: no brackets, no fractions.

**STAGE 2
ISOLATE**

- Collect: x -terms one side (move the **smaller** one)
 - Numbers to the other side
 - Simplify, then divide
- Now: $x = \text{a number}$.

$x = 3$ is the only point that balances $2x + 1 = 7$

**CALCULATOR FIRST, THEN CHECK**

Get the answer first. Type the equation, using **ALPHA** **)** for x and **ALPHA** **CALC** for $=$. Then press **SHIFT** **CALC** (SOLVE) and **=**. Now you know where you are going.

Check at the end. Press **CALC**, type your answer: both sides give the same number? Done.

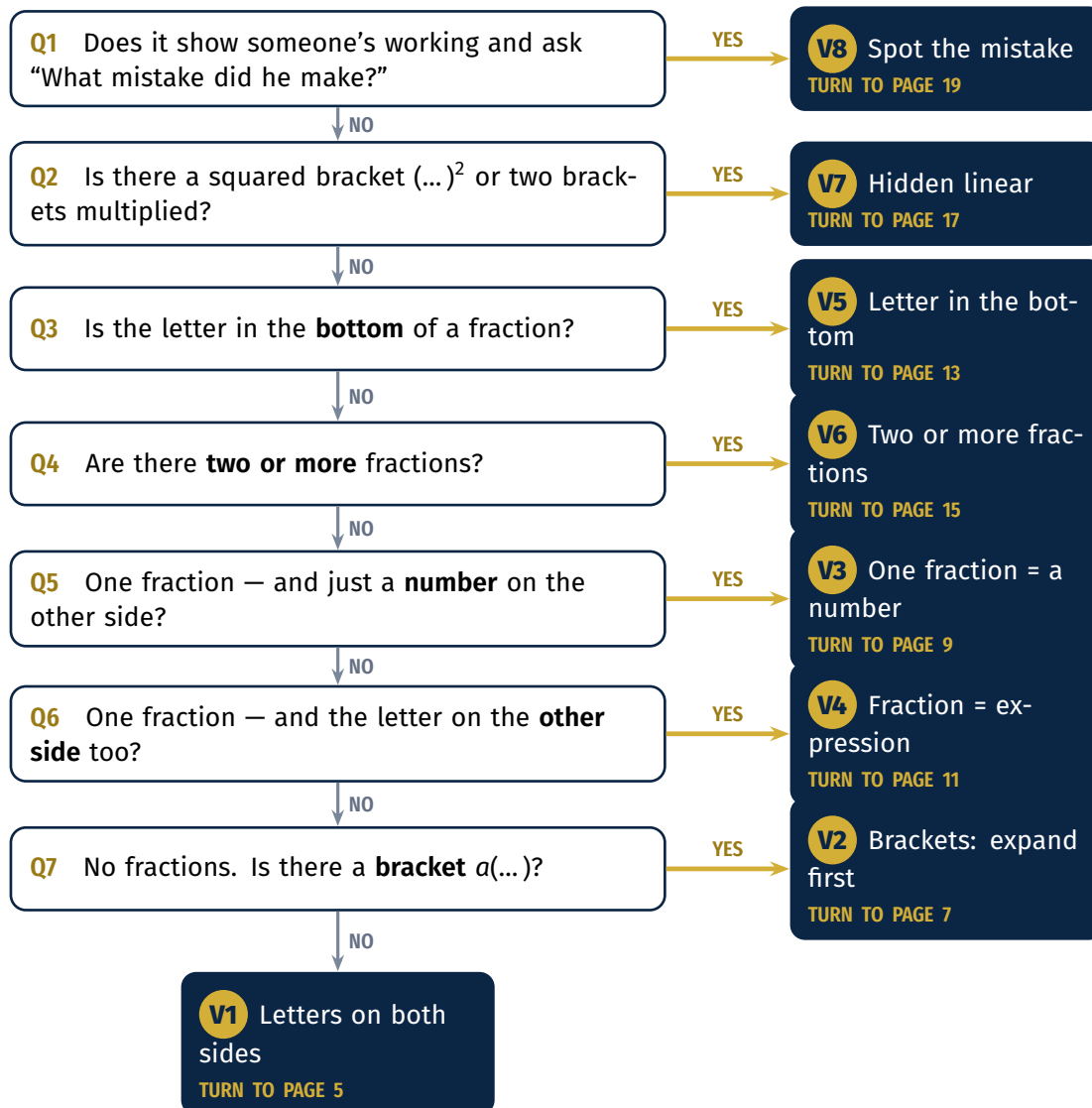
But the exam says **"Show clear algebraic working"** — the calculator answer alone scores 0. Write every line.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3x$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



How do I do it?



Worked example



Check yourself



When you see → you do



Watch out



Exam tip



Calculator check

V1

VARIANT 1

Letters on Both Sides

No brackets, no fractions — just x on the left *and* on the right.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 3


Your notes

? What is it?

IN PLAIN WORDS

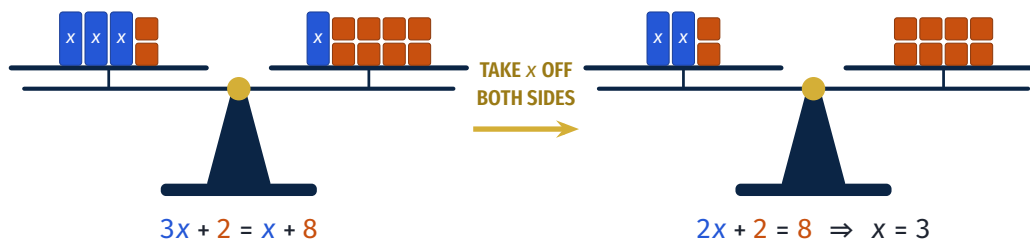
The letter appears on **both sides** of the $=$ sign. Your job: bring all the x 's together on **one** side and all the numbers on the **other** side.

IT LOOKS LIKE

$$5x + 4 = 14 + x$$

$$4x - 13 = 17 + 8x$$

$$10x = 62.4 - 3x$$



Take the same thing off both pans — the scale stays level. Move the **smaller** x -term so the x 's stay positive.

How do I do it?

1 Collect x

Find the **smaller** x -term. Move it across: it **changes sign**.

2 Move numbers

Numbers go to the other side. They change sign too.

3 Simplify

Add up each side: one x -term = one number.

4 Divide

Divide by the number in front of x .

MOVE THE SMALLER x It keeps your x -term positive — fewer minus-sign mistakes.

V1 Letters on Both Sides

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“Solve” and x on **both** sides, no x^2 → It’s linear: collect x on one side $4x$ on the left, $8x$ on the right→ Move the smaller one ($4x$) to the right

“Show clear algebraic working”

→ One line per move — a bare answer can score 0



WORKED EXAMPLE

PRACTICE

Linear Equations · Easy Q9 · 2 marks

Solve $4x - 13 = 17 + 8x$

STEP 1

$$-13 = 17 + 4x$$

◀ $4x$ is smaller than $8x$: take $4x$ off both sides

STEP 2

$$-13 - 17 = 4x$$

◀ take 17 off both sides

STEP 3

$$-30 = 4x$$

◀ simplify

STEP 4

$$x = \frac{-30}{4}$$

◀ divide by 4

ANSWER $x = -7.5$ ✓ CALC $x = -7.5$: both sides give -43 

WATCH OUT

$$\times 4x - 8x = 17 - 13$$

$$\checkmark 4x - 8x = 17 + 13$$

A term that **crosses** the $=$ sign **changes sign**: -13 on the left becomes $+13$ on the right.



EXAM TIP

Get x first with **SHIFT** **CALC**. If your working gives a different number, find the line where the sign flipped.



Check yourself

calculator first · answers on page 21

1.1

Practice · Easy

Solve $5x + 4 = 14 + x$

Answer:

1.2

Practice · Easy

Solve $10x = 62.4 - 3x$

Answer:

1.3

Dr Eslam

Solve $7x - 3 = 2x + 12$

Answer:

V2

VARIANT 2

Brackets: Expand First

A number sits outside a bracket. Clean it away, then it is a V1 equation.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 9


Your notes
? What is it?**In plain words**

$5(x + 3)$ means **5 lots of** $(x + 3)$. The 5 multiplies **every term** inside the bracket.

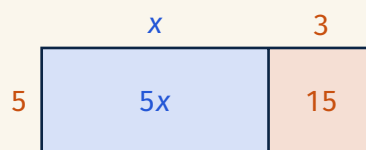
Expand first. After that there is no bracket — just letters and numbers, like V1.

IT LOOKS LIKE

$$3(x - 2) = x + 7$$

$$5(4 - x) = 7 - 3x$$

$$7(k - 3) = 3k - 5$$



$$5(x + 3) = 5x + 15$$

$$5(4 - x) = 20 - 5x$$

Area model and arrows: hit **every** term.

How do I do it?**1 Clean: expand**

Multiply the outside number by **every** term inside. Watch the signs.

2 Collect

Move the smaller x-term across, then the numbers.

3 Simplify

One x-term = one number.

4 Divide

Divide by the number in front of x.

CLEAN → ISOLATE Expanding is Stage 1. Steps 2–4 are exactly V1.

V2 Brackets: Expand First

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

A number in front of a bracket: $5(\dots)$ → **Expand: multiply every term inside**

A minus outside: $7 - 2(x + 1)$ → **The -2 multiplies both: $-2x - 2$**

x on both sides after expanding → **Finish it like V1: move the smaller x**

WORKED EXAMPLE **PRACTICE**

Linear Equations · Easy Q7 · 3 marks

Solve $5(4 - x) = 7 - 3x$. Show clear algebraic working.

STEP 1 $20 - 5x = 7 - 3x$ ◀ expand: 5×4 and $5 \times (-x)$

STEP 2 $20 = 7 + 2x$ ◀ $-5x$ is smaller: add $5x$ to both sides

STEP 2 $20 - 7 = 2x$ ◀ take 7 off both sides

STEP 3 $13 = 2x$ ◀ simplify

STEP 4 $x = \frac{13}{2}$ ◀ divide by 2

ANSWER $x = 6.5$ ✓ CALC $x = 6.5$: both sides give -12.5 **WATCH OUT**

$$\times 5(4 - x) = 20 - x$$

$$\checkmark 5(4 - x) = 20 - 5x$$

The 5 multiplies **both** terms in the bracket — the x too.

EXAM TIP

If the outside number divides everything, you may divide first:
 $7(x + 2) = 7 \Rightarrow x + 2 = 1 \Rightarrow x = -1$.
 Expanding always works.

Check yourself

calculator first · answers on page 21

2.1 Practice · Easy

Solve $4(x + 3) = 2x + 8$

Answer:

2.2 Practice · Medium

Solve $4(3x - 2) = 2x - 5$

Answer:

2.3 Practice · Hard

Solve $5(f - 3) = f + 10$

Answer:



VARIANT 3

One Fraction = a Number

The letter is inside a fraction, and the other side is just a number.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 2



Your notes

What is it?

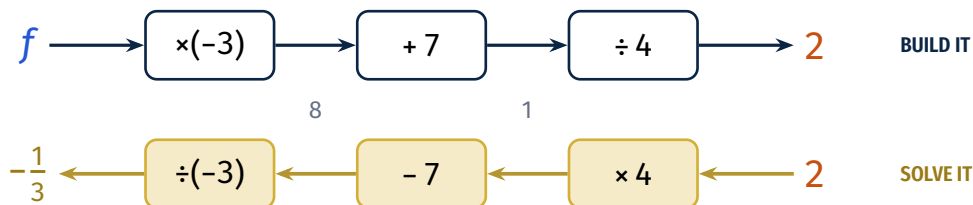
IN PLAIN WORDS

$\frac{7-3f}{4} = 2$ means: something divided by 4 gives 2. So that something must be $2 \times 4 = 8$.

Undo the fraction first: multiply both sides by the bottom number. Then undo the rest, **last thing first**.

IT LOOKS LIKE

$$\frac{2-y}{5} = 1 \quad \frac{7-3f}{4} = 2 \quad \frac{x+5}{3} = 4$$



A function machine: build the equation forwards,
then solve it **backwards** with the opposite operations.

How do I do it?

- 1 Clean: × the bottom**

Multiply both sides by the bottom number. The fraction disappears.

- ## 2 Move the number

Take the plain number away from the x-term (opposite operation).

- ### 3 Divide

Divide by the number in front of x — **keep its sign** if it is negative.

V3 One Fraction = a Number

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

A fraction = a plain number

→ **Multiply both sides by the bottom**A minus x-term on top: $7 - 3f$ → **You will divide by -3: the answer may be negative**An answer like $-\frac{1}{3}$ → **Leave it as a fraction — do not round****WORKED EXAMPLE** **PRACTICE**

Linear Equations • Medium Q4(b) • 3 marks

Solve $\frac{7 - 3f}{4} = 2$

STEP 1 $7 - 3f = 8$

◀ multiply both sides by 4

STEP 2 $-3f = 8 - 7$

◀ take 7 off both sides

STEP 2 $-3f = 1$

◀ simplify

STEP 3 $f = \frac{1}{-3}$

◀ divide by -3 (keep the minus)

ANSWER $f = -\frac{1}{3}$

✓ CALC $f = -\frac{1}{3}$: $\frac{7+1}{4} = 2$

WATCH OUT

$\times -3f = 1 \Rightarrow f = \frac{1}{3}$

✓ $-3f = 1 \Rightarrow f = -\frac{1}{3}$

You divide by -3, not by 3. Positive ÷ negative = negative.

EXAM TIPFunction-machine thinking is fast for these: $2 \times 4 = 8$, $8 - 7 = 1$, $1 \div (-3) = -\frac{1}{3}$. Still write the lines.**Check yourself**

calculator first • answers on page 21

3.1 Practice • Easy

Solve $\frac{2 - y}{5} = 1$

Answer:

3.2 Dr Eslam

Solve $\frac{x + 5}{3} = 4$

Answer:

3.3 Dr Eslam

Solve $\frac{3x - 1}{2} = 7$

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

One Fraction = an Expression

One side is a fraction; the other side has the letter too. The exam's favourite.

SEEN IN PAST PAPERS

●●●●●●●●●● 10 / 12

PRACTICE QUESTIONS: 11



Your notes

Came up in: Jan 2020 P1H Q7(d) • Oct/Nov 2020 P1HR Q3(c) • Jan 2021 P1H Q5(b) • Jan 2022 P2HR Q3(b) • Jun 2022 P1H Q5(b) • Jun 2022 P2HR Q6(a) • May 2024 P1H Q6(b) • Jun 2024 P2HR Q6(a) • Nov 2024 P1H Q3(c) • Jun 2025 P2HR Q1(b)

? What is it?

IN PLAIN WORDS

$\frac{2x+5}{3} = 4-x$ means: **one third** of $(2x+5)$ is $(4-x)$. So the whole of $(2x+5)$ is **three lots of** $(4-x)$: $2x+5 = 3(4-x)$. That bracket is the whole secret.

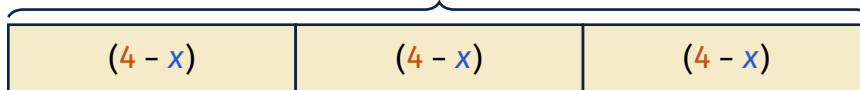
IT LOOKS LIKE

$$\frac{2x+5}{3} = 4-x$$

$$6x-5 = \frac{4x-7}{2}$$

$$p = \frac{3p-5}{10}$$

$$\frac{2f}{3} = 4f-17$$

 $2x+5$ (the top)**3 PARTS**

$$2x+5 = 3(4-x)$$

Bar model: the bottom number tells you how many equal parts. Each part is the other side — **in a bracket**.

How do I do it?

1 Bracket

Put the **other side** in a bracket: $(4-x)$.

2 Multiply

Multiply both sides by the bottom. The fraction is gone:
 $2x+5 = 3(4-x)$.

3 Expand

Multiply out: $3(4-x) = 12-3x$.

4 Collect + divide

Move the smaller x , simplify, divide.

BRACKET THE OTHER SIDE

No bracket $\Rightarrow$ the 3 only hits the 4. That is the No. 1 lost mark.

V4 One Fraction = an Expression

VARIANT 4 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

One fraction = something with x , and
 “Show clear algebraic working”

→ **× the bottom, other side in a bracket**

The letter alone: $p = \frac{3p-5}{10}$

→ $10p = 3p - 5$

A bracket already there:

$$3(2x-5) = \frac{9-x}{2}$$

→ **×2:** $6(2x-5) = 9-x$

WORKED EXAMPLE PAST PAPER

November 2024 • Paper 1H • Q3(c) • 3 marks

Solve $\frac{2x+5}{3} = 4-x$. Show clear algebraic working.

STEP 1+2 $2x + 5 = 3(4-x)$

◀ bracket the other side, then
 multiply both sides by 3

STEP 3 $2x + 5 = 12 - 3x$

◀ expand: 3×4 , $3 \times (-x)$

STEP 4 $2x + 3x = 12 - 5$

◀ collect: x left, numbers right

STEP 4 $5x = 7 \Rightarrow x = \frac{7}{5}$

◀ simplify, divide by 5

ANSWER $x = \frac{7}{5} = 1.4$

✓ CALC $x = 1.4$: both sides give 2.6

WATCH OUT

× $2x + 5 = 12 - x$

✓ $2x + 5 = 12 - 3x$

No bracket: the 3 only hits the 4. With
 the bracket it hits 4 **and** $-x$.

EXAM TIP

10 of the 12 past-paper linear equations
 were this variant (3–4 marks). Write the
 bracket line: it earns a mark.

Check yourself

calculator first • answers on page 21

4.1 Jan 2022 P2HR

Solve $6x - 5 = \frac{4x-7}{2}$

Answer:

4.2 Jun 2022 P2HR

Solve $p = \frac{3p-5}{10}$

Answer:

4.3 Jun 2024 P2HR

Solve $\frac{2f}{3} = 4f - 17$

Answer:

V5

VARIANT 5

Letter in the Bottom

The letter is in the denominator. It is still linear — same move, bigger bottom.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 2


Your notes
? What is it?**In plain words**

The bottom of the fraction has the letter in it, like $3x$ or $x - 1$.

Multiply both sides by the whole bottom.

The letter jumps up out of the bottom and the fraction is gone.

IT LOOKS LIKE

$$\frac{4(8x - 2)}{3x} = 10$$

$$\frac{5}{x - 1} = 2$$

$$\frac{4(8x - 2)}{3x} = 10$$

↓ $\times 3x$ BOTH SIDES

$$4(8x - 2) = 10 \times 3x$$

THE BOTTOM CAN'T BE 0, SO $x \neq 0$

How do I do it?**1 $\times$ the bottom**

Multiply both sides by the **whole** bottom. Two terms?
Bracket it: $2(x - 1)$.

2 Expand

Multiply out every bracket on both sides.

3 Collect + divide

Smaller x across, numbers across, divide.

4 Check

Your x must not make the bottom 0.

V5 Letter in the Bottom

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

The letter in the bottom: $\frac{\dots}{3x} = 10$ → $\times 3x$ **both sides — still linear**

A bottom with two terms: $\frac{5}{x-1} = 2$ → **Bracket it:** $5 = 2(x-1)$

Two fractions with bottoms $3x$ and $2x$ → **LCM = $6x$, then like V6**

WORKED EXAMPLE PRACTICE

Linear Equations · Medium Q2 · 3 marks

Solve $\frac{4(8x-2)}{3x} = 10$

STEP 1 $4(8x-2) = 10 \times 3x$

◀ multiply both sides by $3x$

STEP 2 $32x - 8 = 30x$

◀ expand both sides

STEP 3 $32x - 30x = 8$

◀ collect: x left, number right

STEP 3 $2x = 8 \Rightarrow x = 4$

◀ simplify, divide by 2

STEP 4 $x = 4$ makes the bottom 12, not 0 ✓

◀ check the bottom

ANSWER $x = 4$

✓ CALC $x = 4$: $\frac{4 \times 30}{12} = 10$

WATCH OUT

✗ $4(8x-2) = 10$

✓ $4(8x-2) = 30x$

Multiply **both** sides by $3x$ — the 10 becomes $30x$.

EXAM TIP

A letter in the bottom does not make it a quadratic. After $\times 3x$ look for x^2 : none here, so solve it like V1.

Check yourself

calculator first · answers on page 21

5.1 Dr Eslam

Solve $\frac{5}{x-1} = 2$

Answer:

5.2 Dr Eslam

Solve $\frac{x+6}{2x} = 2$

Answer:

5.3 Practice · Hard

Solve $\frac{x+2}{3x} + \frac{x-2}{2x} = 3$

Answer:

V6

VARIANT 6

Two or More Fractions

Different bottoms. One number clears them all: the LCM.

SEEN IN PAST PAPERS

2 / 12

PRACTICE QUESTIONS: 7



Your notes

Came up in: Jun 2022 P2H Q15(a) • Jun 2025 P1HR Q15(a)

? What is it?

IN PLAIN WORDS

Two (or more) fractions with **different** bottoms. Find the **LCM** of the bottoms — the smallest number they all go into — and multiply **every term** by it. The number on the right too.

IT LOOKS LIKE

$$\frac{5a+8}{3} - \frac{2a+5}{4} = 23$$

$$\frac{4x+5}{3} - \frac{3-2x}{2} = 13$$

$$3(x - 4) + \frac{x + 2}{5} = 6$$

× TABLE OF 3

3 6 9 **12**

LCM = 12

× TABLE OF 4

4 8 **12**

$$\frac{5a+8}{3} - \frac{2a+5}{4} = 23$$

$$\downarrow \times 12 \qquad \downarrow \times 12 \qquad \downarrow \times 12$$

$$4(5a+8) - 3(2a+5) = 276$$

DON'T FORGET ME

Find the LCM, then send it to **every** term. Each bottom cancels and leaves a bracket.

How do I do it?

1 Find the LCM

Smallest number all the bottoms go into. 3 and 4 $\rightarrow$ 12.

2 $\times$ every term

Each bottom cancels. Its top goes in a **bracket**. The right side gets $\times \text{LCM}$ too.

3 Expand

Careful after a minus: $-3(2a + 5) = -6a - 15$.

4 Collect + divide

Simplify each side, then divide.

LCM × EVERY TERM Every term means every term — including the lonely number on the right.

V6 Two or More Fractions

VARIANT 6 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DOTwo fractions with **different** bottoms → **Find the LCM, multiply every term**A minus in front of a fraction → **Bracket its top:** $-3(2a + 5)$ A number alone on the right, = 23 → **Multiply it too:** $23 \times 12 = 276$ **WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 1HR · Q15(a) · 4 marks

Solve $\frac{5a+8}{3} - \frac{2a+5}{4} = 23$. Show clear algebraic working.**STEP 1** LCM of 3 and 4 is 12

◀ find the LCM

STEP 2 $4(5a + 8) - 3(2a + 5) = 276$

◀ multiply every term by 12

STEP 3 $20a + 32 - 6a - 15 = 276$

◀ expand (the minus hits both)

STEP 4 $14a + 17 = 276 \Rightarrow 14a = 259$

◀ simplify, take 17 off

STEP 4 $a = \frac{259}{14}$

◀ divide by 14

ANSWER $a = 18.5$ ✓ CALC $a = 18.5: 33.5 - 10.5 = 23$ **WATCH OUT**✗ $-3(2a + 5) = -6a + 15$ ✓ $-3(2a + 5) = -6a - 15$ The minus in front multiplies **both** terms of the bracket.**EXAM TIP**

Write the bracket line

 $4(5a + 8) - 3(2a + 5) = 276$ first. It earns the method mark even if you slip later.**Check yourself**

calculator first · answers on page 21

6.1

Jun 2022 P2H

Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$

Answer:

6.2

Practice · Hard

Solve $\frac{4x-1}{5} + \frac{x+4}{2} = 3$

Answer:

6.3

Practice · Hard

Solve $3(x-4) + \frac{x+2}{5} = 6$

Answer:

V7

VARIANT 7

Hidden Linear (x^2 Cancels)

It looks like a quadratic. Expand everything — the x^2 terms vanish.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 2

 Your notes

? What is it?

In plain words

There is a squared bracket or two brackets multiplied, so you see x^2 .

Expand **both sides**. If the **same x^2 term** is on both sides, it cancels — what is left is linear.

IT LOOKS LIKE

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$fg(x) = f(x)$$

	$2x$	5
5	$10x$	25
$2x$	$4x^2$	$10x$

$$(2x + 5)^2 = 4x^2 + 20x + 25$$

DON'T LOSE THE TWO MIDDLE PIECES

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

EXPAND BOTH SIDES

$$4x^2 + 20x + 25 = 4x^2 + 4x - 3$$

SAME x^2 ON BOTH SIDES: CANCEL

$$20x + 25 = 4x - 3 \quad \text{LINEAR!}$$

Expand both sides, then the same x^2 disappears from both — a linear equation is left.

How do I do it?

1 Expand both sides

Square = bracket $\times$ itself. Multiply out every bracket.

2 Cancel the x^2

The same x^2 term on both sides: take it off both.

3 Solve the linear

Collect, simplify, divide — like V1.

V7 Hidden Linear (x^2 Cancels)

VARIANT 7 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

$(\dots)^2$ on one side, $(\dots)(\dots)$ on the other → **Expand both — the x^2 should cancel**

"Solve $fg(x) = f(x)$ " → **Build $f(g(x))$ first, then expand**

$(2x + 5)^2$ → **$(2x + 5)(2x + 5)$ — four pieces, never $4x^2 + 25$**

WORKED EXAMPLE **PRACTICE**

Linear Equations · Hard Q7 · 3 marks

Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

STEP 1 $4x^2 + 20x + 25 = 4x^2 + 4x - 3$ ◀ expand both sides

STEP 2 $20x + 25 = 4x - 3$ ◀ $4x^2$ on both sides: cancel

STEP 3 $20x - 4x = -3 - 25$ ◀ collect

STEP 3 $16x = -28 \Rightarrow x = \frac{-28}{16}$ ◀ simplify, divide by 16

ANSWER $x = -\frac{7}{4}$ ✓ CALC $x = -1.75$: both sides give 2.25**WATCH OUT**✗ $(2x + 5)^2 = 4x^2 + 25$ ✓ $(2x + 5)^2 = 4x^2 + 20x + 25$

Squaring a bracket means bracket × bracket. Look at the area square: two middle pieces of $10x$.

EXAM TIP

If the x^2 terms do **not** cancel, it is a real quadratic — go to Topic 10 (Solving Quadratics).

Check yourself

calculator first · answers on page 21

7.1 Practice · Hard

$f(x) = x^2 + 6$,
 $g(x) = x - 10$.
 Solve $fg(x) = f(x)$

Answer:

7.2 Dr EslamSolve $(x + 3)^2 = x^2 + 21$

Answer:

7.3 Dr Eslam

Solve
 $(x + 5)(x - 1) = (x + 1)^2$

Answer:

V8

VARIANT 8

Spot the Mistake

Someone else's working, one error. Be the examiner.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 1



Your notes

? What is it?

In plain words

The question shows a student's working and says the answer is wrong.

Find the **first wrong line** and say **what it should be**. Usually 1 mark — one clear sentence.

IT LOOKS LIKE

"What mistake did he make?"

	$5(x + 2) = 47$
	$5x + 2 = 47$ ← $5 \times 2 = 10$
	$5x = 45$
	$x = 9$ ✗

How do I do it?

1 Solve it yourself

Quickly, on the side. Now you know the right answer.

2 Compare line by line

Find the **first** line that is different from yours.

3 Say it

What he did, and what it should be: " $5x+2$ should be $5x+10$."

V8 Spot the Mistake

VARIANT 8 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“What mistake did he make?”

→ **Name the line, give the correct version**

“Steve’s answer is wrong.”

→ **Look for the first wrong line — the rest just follows**

(1 mark)

→ **One clear sentence with the correct line****WORKED EXAMPLE** **PRACTICE**

Linear Equations • Easy Q6 • 1 mark

Steve solves $5(x + 2) = 47$. His working: $5x + 2 = 47$, $5x = 45$, $x = 9$.
Steve’s answer is wrong. What mistake did he make?

STEP 1 $5x + 10 = 47 \Rightarrow 5x = 37 \Rightarrow x = 7.4$ ◀ solve it yourself first**STEP 2** Line 1: he wrote $5x + 2$, you have $5x + 10$ ◀ compare line by line**STEP 3** “He did not multiply the 2 by 5. It should be $5x + 10 = 47$.” ◀ say what it should be**ANSWER** $5(x + 2) = 5x + 10$, not $5x + 2$ ✓ then $x = 7.4$ **WATCH OUT**

✗ “He got the wrong answer.”

✓ “ 5×2 should be 10: it is $5x + 10$.”

Vague answers score 0. Point to the exact line and write the correct version.

EXAM TIP

The mistake is almost always one of the Watch-outs in this book: a missing bracket, a sign that did not flip, or a term not multiplied.

Check yourself

calculator first • answers on page 21

8.1

Dr Eslam

Ali: $3(x - 4) = 2x + 1$
 $\rightarrow 3x - 4 = 2x + 1 \rightarrow x = 5$.
 What mistake did he make?

Answer:

8.2

Dr Eslam

Mia: $\frac{x+2}{3} = x - 4$
 $\rightarrow x + 2 = 3x - 4 \rightarrow x = 3$.
 What mistake did she make?

Answer:

8.3

Dr Eslam

Sam: $4x + 3 = 2x - 7$
 $\rightarrow 6x = -10 \rightarrow x = -\frac{5}{3}$.
 What mistake did he make?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 $x = 2.5$

1.2 $x = 4.8$

1.3 $x = 3$

2.1 $x = -2$

2.2 $x = 0.3$

2.3 $f = 6.25$

3.1 $y = -3$

3.2 $x = 7$

3.3 $x = 5$

4.1 $x = \frac{3}{8}$

4.2 $p = -\frac{5}{7}$

4.3 $f = 5.1$

5.1 $x = 3.5$

5.2 $x = 2$

5.3 $x = -\frac{2}{13}$

6.1 $x = 5.5$

6.2 $x = \frac{12}{13}$

6.3 $x = 5.5$

7.1 $x = 5$

7.2 $x = 2$

7.3 $x = 3$

8.1 Line 1:
 $3 \times (-4) = -12$, so
 $3x - 12 = 2x + 1$; $x = 13$

8.2 Line 1: the bracket
 is missing,
 $x + 2 = 3(x - 4) = 3x - 12$;
 $x = 7$

8.3 Line 1: take $2x$ off,
 do not add it: $2x = -10$;
 $x = -5$

THE WHOLE TOPIC ON ONE PAGE

Linear Equations cheat sheet



Your notes
CLEAN

brackets, fractions

COLLECT

move the smaller x

SIMPLIFY

one term each side

DIVIDE

then CHECK with CALC

V1 Letters on both sides

$$4x - 13 = 17 + 8x$$

Move the **smaller** x-term. Crossing = flips the sign.

V2 Brackets

$$5(4 - x) = 7 - 3x$$

Expand: the outside number hits **every** term. Then V1.

V3 Fraction = number

$$\frac{7-3f}{4} = 2$$

× the bottom. Undo last-thing-first. Keep the minus.

V4 Fraction = expression 10/12

$$\frac{2x+5}{3} = 4 - x$$

Bracket the other side, then × the bottom:
 $2x + 5 = 3(4 - x)$.

V5 Letter in the bottom

$$\frac{4(8x-2)}{3x} = 10$$

× the **whole** bottom. Check it isn't 0.

V6 Two fractions 2/12

$$\frac{5a+8}{3} - \frac{2a+5}{4} = 23$$

LCM × **every** term — the right side too.
 Bracket after a minus.

V7 Hidden linear

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

Expand both sides; the same x^2 cancels.
 $(a + b)^2$ has 4 pieces.

V8 Spot the mistake

$$5x + 2 = 47 \times$$

Solve it yourself, find the **first** wrong line, write the correct line.

**TOP 3 LOST MARKS**

1. No bracket: $2x + 5 = 12 - x \times$
2. Minus before a bracket: $-6a + 15 \times$
3. Right side not multiplied: $= 23 \times$

**CALCULATOR FIRST, THEN CHECK**

Type it: **ALPHA** **)** = x, **ALPHA** **CALC** = “=”.
 Solve: **SHIFT** **CALC** **=**. Check: **CALC**.
 Then write **every** line — “Show clear algebraic working”.

BOTTOM LINE Clean → Collect → Simplify → Divide → Check. Fractions?
Bracket first.